California Coastal Waters

What 70 Years of Ocean Data Reveals About Temperature and Dissolved Oxygen

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The California Cooperative Oceanic Fisheries Investigations (CalCOFI) program has been collecting water samples off the California coast since 1949. The dataset spans more than 70 years and nearly 900,000 individual water bottle samples, making it one of the most complete long-term ocean monitoring records available. This project examines two questions: whether water temperature and dissolved oxygen levels have changed over that period, and whether the two variables are related.

Dissolved oxygen matters because most marine life depends on it. Fish, invertebrates, and other organisms need adequate oxygen to survive. When oxygen levels drop too low, usually below 1.4 ml/L (equivalent to the standard 2 mg/L hypoxia threshold), water becomes hypoxic and most species either leave or die (Diaz & Rosenberg, 2008). Understanding what drives oxygen availability, and whether it is changing over time, has direct implications for fisheries management and coastal ecosystem health.

Data and Methods

The analysis began with two raw datasets covering bottle samples and cast metadata, which were merged on a shared cast identifier to bring date and location into the same working dataset. After cleaning, the final dataset contained 720,707 usable records. Temperature ranged from 1.44 to 31.14 degrees Celsius, dissolved oxygen from 0 to 11.13 ml/L, and depth from the surface down to 5,351 meters. Missing salinity values were filled using the median rather than the mean, since freshwater input from rivers and coastal runoff can produce low salinity outliers that would skew a mean calculation (Bruce et al., 2020).

Findings

The univariate analysis showed a bimodal distribution of dissolved oxygen. Rather than a single bell curve, the data showed two distinct clusters: one near 0.5 ml/L representing deep water samples, and a second around 6 ml/L representing shallower surface water. This indicated that depth would be a factor throughout the rest of the analysis.

The bivariate analysis confirmed a strong positive correlation of 0.80 between temperature and dissolved oxygen. This result is counterintuitive, since warmer water physically holds less oxygen. Depth explains the discrepancy. Shallow water is warmer and oxygen-rich due to wave mixing and photosynthesis. Deep water is colder and oxygen-depleted because it is cut off from both. Depth drives both variables in the same direction, producing the positive correlation. A

Pearson correlation test returned a p-value of essentially zero, confirming the relationship is statistically significant.

Regression modeling produced consistent results. A linear regression using temperature to predict oxygen produced an R^2 of 0.64, meaning temperature alone explains about 64% of the variation in oxygen. A polynomial regression improved that to 0.70, better capturing the curved shape visible in the scatter plot. The rest is mostly explained by depth, which was not included as a predictor in either model.

Logistic Regression classified whether a sample had high or low oxygen (relative to the median) with 94.8% accuracy using only temperature and salinity as predictors. A Random Forest Classifier improved that slightly to 95.0%. The small gap between the two makes sense given that the relationship is already mostly linear.

K-means clustering produced the clearest structural result. Using temperature, oxygen, and salinity as inputs, the algorithm grouped the data into three clusters that mapped almost exactly onto known oceanographic water mass types: warm shallow surface water with high oxygen, a mid-depth transitional layer, and cold deep water with near-zero oxygen. A silhouette score of 0.44 means the clusters are moderately well-separated. That's reasonable since ocean water masses don't have sharp edges and tend to blend into each other. The algorithm had no labels or guidance and found these groupings on its own.

The time trend analysis shows how annual average temperature and oxygen have moved together from 1949 through 2021. Before 1970, both variables show considerable year-to-year noise, likely reflecting inconsistent sampling in the early decades of the program. From the 1980s onward they track each other closely. Both show a declining trend through 2021, consistent with known climate patterns during that period.

Recommendations

Two recommendations follow from these findings. First, depth should be incorporated as a predictor variable in future regression models. The analysis repeatedly showed depth acting as the underlying driver connecting temperature, oxygen, and salinity. Leaving it out limits predictive power.

Second, the time trend data warrants continued monitoring. The declining trend in both temperature and oxygen through 2021 is consistent with known climate patterns, but sustained declines in oxygen would have serious consequences for the coastal ecosystems NOAA monitors at sites like Monterey Bay. On a practical level, the classification model's high accuracy suggests that temperature and salinity readings alone could flag low-oxygen conditions in coastal monitoring, without requiring direct oxygen measurement at every sampling point.

References

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